

Mathematics A

Unit 2 • Chapter OL-T42 Classified

Cross-session • chapter-organised candidate

16 classified items • 60 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Unit 2 • Chapter OL-T42 • 16 items • 60 marks

Chapter	Items	Marks	Starts
01. OL-T42 Circle Theorems	16	60	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Circle Theorems

Unit 2 • 60 marks • 16 item(s)

June 2024 | 4MA1-2H Q21 | 5 marks

Use radius perpendicular to tangent

November 2023 | 4MA1-1H Q14 | 3 marks

Complete a multi-theorem circle-angle chain

November 2023 | 4MA1-1H Q22 | 6 marks

Use radius perpendicular to tangent

June 2023 | 4MA1-2H Q7 | 3 marks

Use radius perpendicular to tangent

June 2023 | 4MA1-2H Q14 | 4 marks

Use angles in the same segment

November 2024 | 4MA1-2H Q16 | 5 marks

Complete a multi-theorem circle-angle chain

January 2022 | 4MA1-2H Q16 | 4 marks

Complete a multi-theorem circle-angle chain

January 2022 | 4MA1-2H Q19 | 3 marks

Use the internal intersecting-chords product

January 2020 | 4MA1-2H Q13 | 4 marks

Complete a multi-theorem circle-angle chain

January 2021 | 4MA1-2H Q13 | 2 marks

Complete a multi-theorem circle-angle chain

January 2021 | 4MA1-2H Q13 | 1 marks

Complete a multi-theorem circle-angle chain

June 2021 | 4MA1-1H Q15 | 4 marks

Complete a multi-theorem circle-angle chain

November 2021 | 4MA1-2H Q11 | 3 marks

Complete a multi-theorem circle-angle chain

May-Nov 2020 | 1H Q16 | 4 marks

Complete a multi-theorem circle-angle chain

January 2023 | 2H Q14 | 5 marks

Complete a multi-theorem circle-angle chain

... 1 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Question 24

4MA1-2H Q21

MODULAR UNIT 2

5 marks

- 21 The diagram shows a square $ABCD$ and a circle.

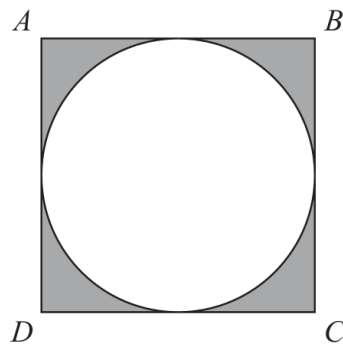


Diagram **NOT**
accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is 80 cm^2

Work out the length of AC

Give your answer correct to 3 significant figures.

..... cm

WORKING SPACE

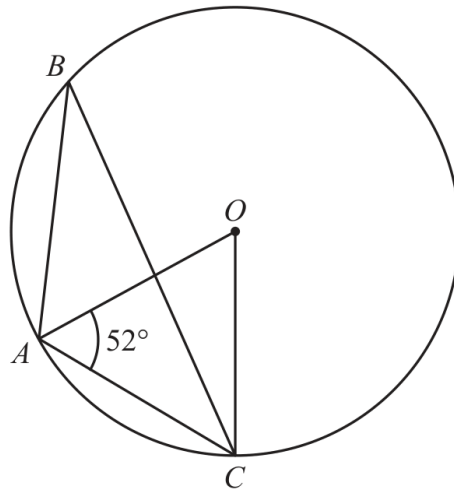
Question 11

4MA1-1H Q14 demand

MODULAR UNIT 2

3 marks

14

Diagram **NOT**
accurately drawn

A , B and C are points on a circle, centre O

Angle $OAC = 52^\circ$

Find the size of angle ABC

Give reasons for your working.

○

WORKING SPACE

Working space for Question 11

Continue your method

UNIT 2

3 marks

WORKING SPACE

Question 17

4MA1-1H Q22 demand

MODULAR UNIT 2

6 marks

- 22 [In this question 1 cm = 1 unit on the x -axis and
1 cm = 1 unit on the y -axis]

P is a point on a circle with centre $(0, 0)$

The coordinates of P are $(8, -10)$

The line L is the tangent to the circle at the point P

L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ

Give your answer correct to 3 significant figures.

..... cm

WORKING SPACE

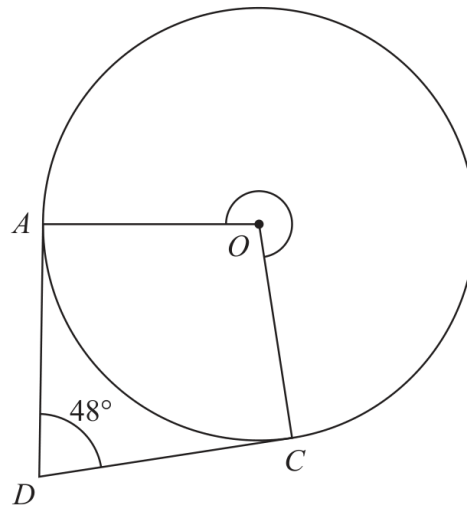
Question 21

4MA1-2H Q7

MODULAR UNIT 2

3 marks

7

Diagram **NOT**
accurately drawn

A and C are points on a circle, centre O

DA is the tangent to the circle at A and DC is the tangent to the circle at C

Angle $ADC = 48^\circ$

Work out the size of reflex angle AOC

.....
○

Working space for Question 21

Continue your method

UNIT 2

3 marks

WORKING SPACE

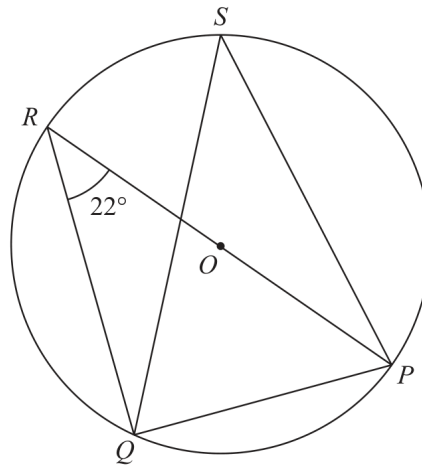
Question 25

4MA1-2H Q14

MODULAR UNIT 2

4 marks

14

Diagram **NOT**
accurately drawn

P , Q , R and S are points on a circle, centre O
 ROP is a diameter of the circle.
 Angle $PRQ = 22^\circ$

(a) (i) Find the size of angle RQP

.....
 (1)

(ii) Give a reason for your answer.

.....

 (1)

(b) (i) Find the size of angle PSQ

.....
 (1)

(ii) Give a reason for your answer.

.....

 (1)

Working space for Question 25

Continue your method

UNIT 2

4 marks

WORKING SPACE

Question 28

4MA1-2H Q16 M01, M02, M03, M04

MODULAR UNIT 2

5 marks

16 A, B, C and D are points on a circle, centre O

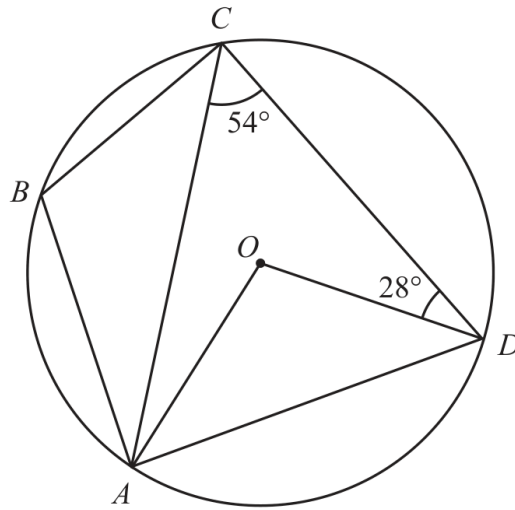


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle AOD

.....
(1)

(ii) Give a reason for your answer to part (a)(i)

.....
.....
(1)

(b) Work out the size of angle CAO

.....
(1)

(c) Work out the size of angle ABC

.....
(2)

Working space for Question 28

Continue your method

UNIT 2
5 marks

WORKING SPACE

Question 29

4MA1-2H Q16

MODULAR UNIT 2

4 marks

16 D, E, F and G are points on a circle, centre O

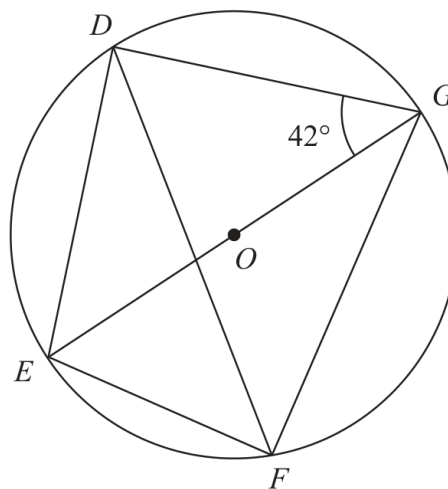


Diagram **NOT**
accurately drawn

EOG is a diameter of the circle.

Angle $EGD = 42^\circ$

Calculate the size of angle DFG

Give a reason for each stage of your working.

Angle $DFG = \dots\dots\dots^\circ$

Working space for Question 29

Continue your method

UNIT 2

4 marks

WORKING SPACE

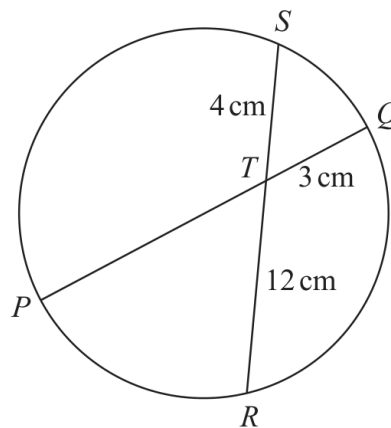
Question 31

4MA1-2H Q19

MODULAR UNIT 2

3 marks

19

Diagram **NOT**
accurately drawn PTQ is a diameter of a circle. RTS is a chord of the circle.

$TQ = 3\text{ cm}$

$ST = 4\text{ cm}$

$TR = 12\text{ cm}$

Calculate the radius of the circle.

..... cm

Working space for Question 31

Continue your method

UNIT 2

3 marks

WORKING SPACE

13

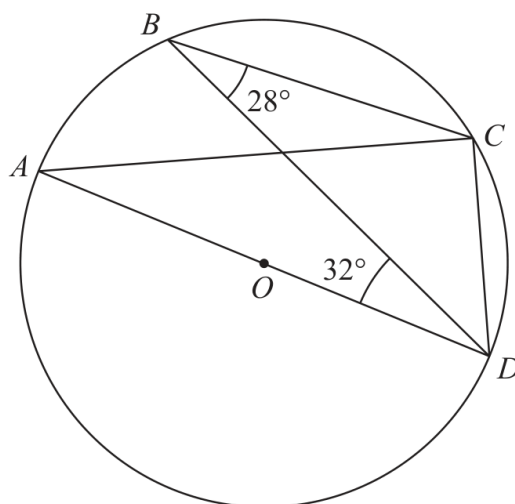


Diagram **NOT**
accurately drawn

A , B , C and D are points on a circle, centre O .
 AOD is a diameter of the circle.

Angle $CBD = 28^\circ$

Angle $BDA = 32^\circ$

Find the size of angle BDC .

Give a reason for each stage of your working.

YOUR WORKING

13 P, Q, R, S and T are points on a circle with centre O .

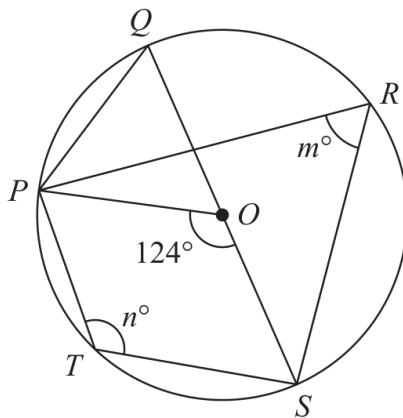


Diagram **NOT**
accurately drawn

QOS is a diameter of the circle.

angle $POS = 124^\circ$ angle $PRS = m^\circ$ angle $PTS = n^\circ$

(a) Find the value of

(i) m

(ii) n

(b) Find the size of angle QPO .

.....

.....

(2)

.....

(1)

YOUR WORKING

13 P, Q, R, S and T are points on a circle with centre O .

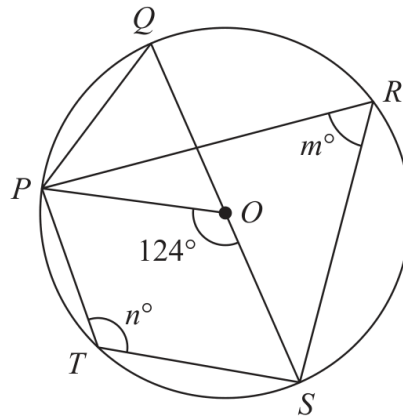


Diagram **NOT**
accurately drawn

QOS is a diameter of the circle.

angle $POS = 124^\circ$ angle $PRS = m^\circ$ angle $PTS = n^\circ$

(a) Find the value of

(i) m

(ii) n

(b) Find the size of angle QPO .

.....

.....

(2)

.....

(1)

YOUR WORKING

- 15 P , Q and R are points on a circle, centre O .
 TRV is the tangent to the circle at R .

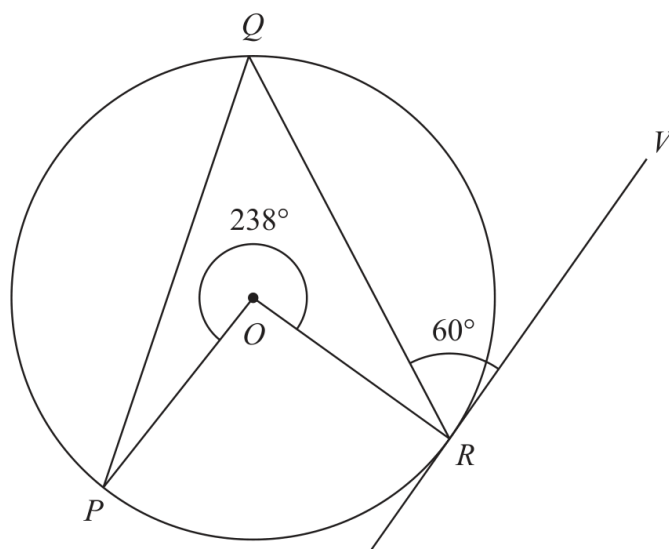


Diagram **NOT**
 accurately drawn

Reflex angle $POR = 238^\circ$

Angle $QRV = 60^\circ$

Calculate the size of angle OPQ .

Give a reason for each stage of your working.

YOUR WORKING

11

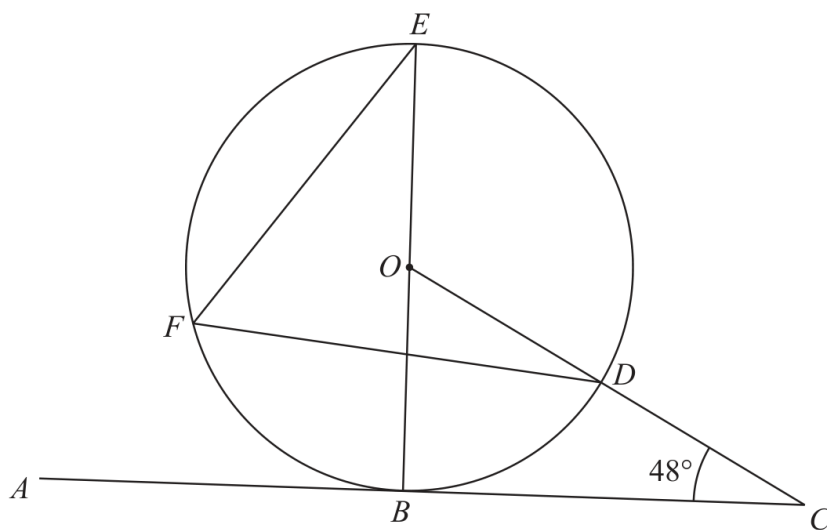


Diagram **NOT**
accurately drawn

B, D, E and F are points on a circle, centre O .

ABC is a tangent to the circle.

ODC is a straight line.

BOE is a diameter of the circle.

Angle $BCD = 48^\circ$

Find the size of angle DFE .

YOUR WORKING

16

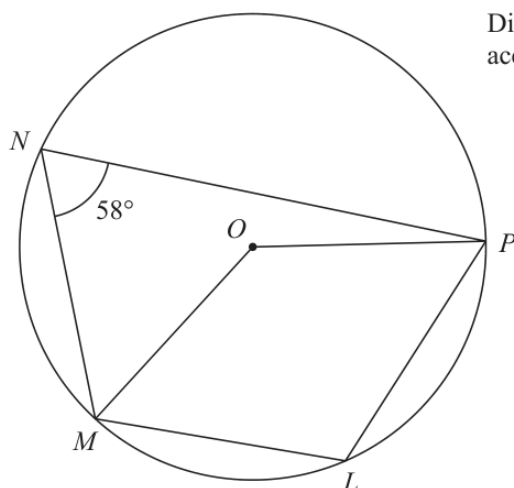


Diagram **NOT**
accurately drawn

L, M, N and P are points on a circle, centre O

Angle $MNP = 58^\circ$

(a) (i) Find the size of angle MLP

o

(ii) Give a reason for your answer.

(2)

(b) Find the size of the reflex angle MOP

o

(2)

(Total for Question 16 is 4 marks)

YOUR WORKING

14 A, B, C and D are points on a circle, centre O

EBF is the tangent to the circle at B

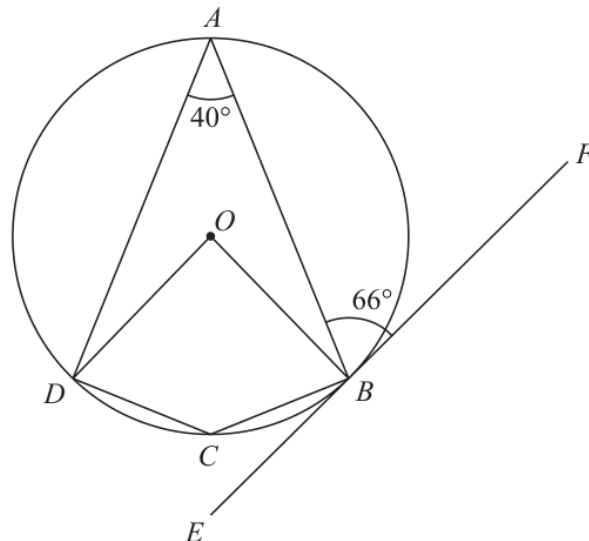


Diagram **NOT** accurately drawn

(a) (i) Work out the size of angle DCB

.....
(1)

(ii) Give a reason for your answer to (a)(i)

.....
.....
(1)

(b) Work out the size of angle ADO

.....
(3)

(Total for Question 14 is 5 marks)

YOUR WORKING

15 A, B, C and D are points on a circle with centre O

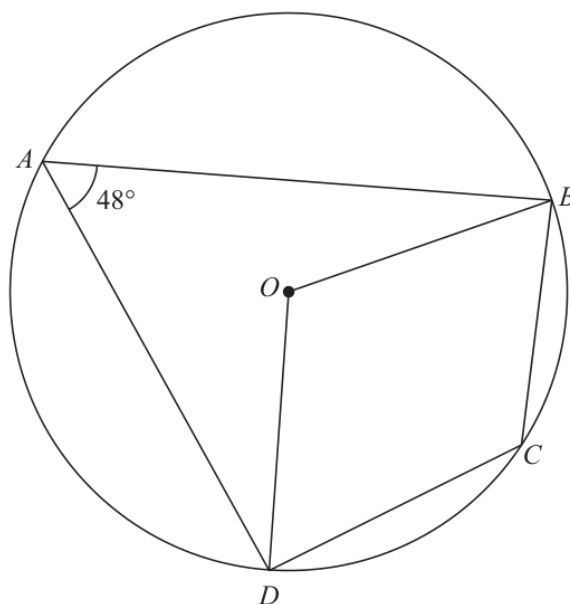


Diagram **NOT** accurately drawn

Angle $DAB = 48^\circ$

(a) (i) Work out the size of the obtuse angle DOB

(1)

(ii) Give a reason for your answer.

(1)

(b) (i) Work out the size of angle BCD

(1)

(ii) Give a reason for your answer.

(1)

(Total for Question 15 is 4 marks)

YOUR WORKING